Ashutosh Sharma

Software Engineering · Cloud · DevOps · Backend · Data · Python · Java · AWS

Gurgaon, Haryana · +91 96501 64430 · 9sharmaa@gmail.com · linkedin.com/in/ashutosharma2 · github.com/ashutosharma2

SUMMARY

IT undergraduate who builds software end to end, from React screens and Python APIs to the AWS and Kubernetes infrastructure under them, and tests it by breaking it. Built a serverless app on AWS with all 25 resources in Terraform and 201 tests, a FastAPI and PostgreSQL booking system that never double-booked through fault drills, and a Python pipeline that replicates a published study on a 29 GB dataset. Seeking a software, cloud, DevOps, backend or data internship.

EDUCATION

B.Tech, Information Technology — Netaji Subhas University of Technology (NSUT), New Delhi · Expected 2028
Relevant coursework: Data Structures and Algorithms, Advanced Programming (Java, grade 9/10), Software Development (Agile, Scrum)
LeetCode: 64 solved (33 Easy, 24 Medium, 7 Hard) · NeetCode: 35 problems in Java

Class XII, High School Diploma — American International School of Mozambique (AISM), Maputo · 2023 · 89%

PROJECTS

LiftLens — full-stack physique tracker on AWS · React, TypeScript, Python, AWS Lambda, DynamoDB, Cognito, Terraform · GitHub

- Built a 3-tier serverless web app (React/TypeScript, Python REST API on AWS Lambda, DynamoDB, Cognito) with all 25 AWS resources in Terraform across three stacks; the API and data are live, and app hosting on CloudFront is next.
- Implemented JWT verification against Cognito, SSO with no long-lived keys and least-privilege IAM roles with no wildcard actions; ran pose estimation in the browser so photos never leave the device.
- Developed 112 Python and 89 TypeScript tests with test doubles, including forged-signature and expired-token cases; a GitHub Actions pipeline runs lint, tests, the build and Terraform validation on every pull request, with actions pinned to commit SHAs.
- Achieved a $0 monthly cost with DynamoDB on provisioned capacity inside the free tier, with CloudWatch logs, budget alerts through SNS, and a kill-switch Lambda and deploy lock that trigger if spending passes $1, tested end to end.

SwimSync — booking system that never oversells · Python, FastAPI, PostgreSQL, React, Docker, Kubernetes, Terraform · GitHub

- Engineered overselling protection under concurrency: a PostgreSQL row lock makes the capacity check and the booking one transaction, backed by a CHECK constraint and a partial unique index; race tests with 20–30 simultaneous requests fail 5 out of 5 times without the lock.
- Implemented safe retries with idempotency keys and a transactional outbox, and diagnosed a lock hang with `pg_stat_activity`; 60 backend tests run against real PostgreSQL in CI, plus 37 frontend tests.
- Deployed to a local Kubernetes cluster that a five-job GitHub Actions pipeline builds and books through on every push; eliminated requests dropped during rolling restarts with a 5 s preStop pause (4 errors in 308 requests before, none in 291 after).
- Conducted three fault drills (instance killed behind HAProxy; database cut or slowed with Toxiproxy; pods force-killed), all with zero duplicate or false bookings, each written up with root cause and fix; designed the AWS deployment (VPC, ALB, EC2, RDS) in Terraform.
- Added monitoring with Prometheus and Grafana: the API exports latency histograms and booking-outcome counters, a dashboard and two alert rules are kept as code, and the counters matched a load test's 3,809 attempts exactly; the outbox alert fired 70 s after the worker was stopped.

SpikeCheck — replication of a cloud forecasting paper · Python, pandas, NumPy, PyTorch, matplotlib, Bash · GitHub

- Built a pipeline that streams a 29 GB cluster trace into 5-minute CPU series for 5,946 containers and trains models in PyTorch; float16 memory-mapped inputs cut peak memory from 3.9 GB to 2.5 GB, so training fits an 8 GB laptop.
- Identified two flawed metrics in a 2025 *Future Generation Computer Systems* paper, confirmed in the authors' code, and showed a simpler approach matches its spike recall with 14 to 29% less over-provisioning across 3 seeds with no actuation delay; with a 15-minute delay, the paper's method wins.

EXPERIENCE

Digital Marketing Intern — Socialbent, Gurgaon · May 2025 – Jul 2025

- Researched markets, competitors and users for client campaigns, alongside SEO research and content optimization.
- Monitored campaign performance with the marketing team as part of campaign management.

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash, HTML, CSS
Frontend and backend: React, Next.js, Vite, Tailwind CSS, FastAPI, REST APIs, JWT authentication
Data: PostgreSQL, DynamoDB, pandas, NumPy, PyTorch, matplotlib, Microsoft Excel
Cloud and DevOps: AWS (Lambda, DynamoDB, Cognito, IAM, CloudWatch, SNS; VPC, ALB, EC2, RDS in Terraform), Terraform, Docker, Kubernetes (kind), GitHub Actions, Prometheus, Grafana, nginx, HAProxy, Linux
Testing and practices: pytest, Vitest, test doubles, Git and pull requests, fault injection, incident write-ups, Agile and Scrum

CERTIFICATIONS

NPTEL Cyber Security and Privacy (IIT Madras) — Elite + Gold, 93%, top 1% of 20,949 candidates · AWS Certified Cloud Practitioner — in preparation · AWS Certified Solutions Architect – Associate — planned